

A competition analysis of the single and double disciplines in world-class badminton.

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Abstract

Competition analyses in badminton commonly focus on the singles disciplines while relatively few studies deal with the demands of the doubles disciplines. Knowledge about the characteristics of the doubles disciplines, however, is essential for coaches in elite badminton in order to set up specific endurance training. For this purpose, the time structure of 50 matches of the world's top ten players in the singles and doubles was investigated. The lowest work to rest ratios were observed in the doubles disciplines (approximately 20 %) except for the ladies' doubles with 30.1 %. Furthermore, the ladies' doubles consisted of the longest rallies on average (10.1 s) whilst the rest times did not differ significantly between the disciplines. The men's and mixed doubles were characterized by the shortest mean rally times of 5.6-6.7 seconds and the highest number of shots per second. These empirical data confirmed the assumed differences between the ladies' doubles and the other doubles disciplines. When setting up specific training drills the higher frequencies in the doubles disciplines should be taken into consideration. The most common work intervals of 6-10 seconds should be applied and drills with the longest possible work intervals should be incorporated into the specific endurance training.

Keywords: competition analysis, racket sports, singles and doubles, world-class badminton

1. Introduction

To the observer of a world-class ladies' double the rallies sometimes seem surprisingly long. The ability of the female players to defend the opponent's attacks are equal to their male counterparts, but the velocity of their offensive strokes is about 13 % lower (Gawin, Beyer, Büsch & Høi, 2012). As a consequence the smashes are often returned successfully and the shuttle seldom seems to touch the ground. Notwithstanding these observations, however, there is a lack of empirical data about the characteristics of the

doubles disciplines in badminton, especially in the field of ladies' or mixed doubles. To date only a few publications have focused on doubles games or have compared singles and doubles games (e.g. Alcock & Cable, 2009; Liddle, Murphy & Bleakley, 1996; Liddle & O'Donoghue, 1998), whilst many more studies deal with the singles disciplines (e.g. Abian-Vicen, Castanedo, Abian & Sampedro, 2013; Abián, Castanedo, Feng, Sampedro & Abian-Vicen, 2014; Cabello, Padial, Lees & Rivas, 2004; Faude, Meyer, Rosenberger, Fries, Huber & Kindermann, 2007; Lee, Xie & Teh, 2008; Leser & Oswald, 2009; Majumdar, Khanna, Malik, Sachdeva, Arif & Mandal, 1997; Tu, 2007).

Aside from the lack of empirical data it needs to be considered that the scoring system has been modified more than once over the years by the introduction of the rally point system. The "traditional" system was a 3 games to 15 format (with the exception of the ladies' singles) and a player or team could only score when being in charge of the service. This scoring system underwent a transition to the 5 games to 7 format combined with the rally point system, which was modified again in 2006 to the 3 games to 21 format, also under rally point scoring conditions. These changes are likely to have affected the game structure, albeit with inconsistent outcomes. According to Tu (2007), while significant differences in the total match duration could not be found, the average performance time in men's singles was reduced while this parameter increased in ladies' singles. By contrast, Chen and Chen (2008) revealed significantly lower values of the analysed temporal structure variables (i.e. total match duration, performance time and rest intervals) in matches with the rally point regulations. These divergent findings were explained on account of the higher skill level of the subjects in Tu's (2007) study of participants in world-class tournaments, while Chen and Chen (2008) analysed Taiwanese national-ranked players. Irrespective of this inconsistency, the results of analyses under the "traditional" scoring system cannot easily be compared with the outcome of studies under current scoring conditions.

Finally, it is noteworthy that many of the aforementioned studies are based on simulated competitions and not on matches that took place in real world-series tournaments (e. g. Alcock & Cable, 2009; Faude et al., 2007; Liddle et al., 1996; Majumdar et al., 1997) and it cannot be assumed that a match under training conditions will generate a time structure identical to a match in a real tournament.

Over and above the particular research conditions, the skill level of the analysed players appears to affect the results of the existing studies. For example, the parameters of rally and rest time are characterized by a broad range according to the discipline and the skill level of the participants (see Table 1). In men's singles average rally times from 5.0-10.4 s are reported. The rest times differ even more with 8.1-26.7 s. In this parameter internationally ranked top players evidently use longer resting periods between the rallies than players at the national level (see Table 1).

Knowledge of the time structure across all disciplines is essential for coaches in elite badminton. The intermittent activity of the game with work intervals of high intensity and relatively short rest times causes high physiological strain for world-class players (Cabello Manrique & González-Badillo, 2003). Therefore discipline-specific data about

the structure of the repeated sequences of work and rest are required for the coaches to set up specific training drills (Liddle & O'Donoghue, 1998).

In view of these considerations, there is a lack of information concerning the game structure (a) in the doubles disciplines in general and especially in the ladies' and mixed doubles, (b) of world-class players in real tournament settings, and (c) under the current scoring conditions.

In the present research, therefore, matches of the world's top ten singles and doubles players were analysed to obtain specific data of the game structure and to compare the singles and doubles disciplines with the aim to develop practical guidelines or recommendations for coaches and players in elite badminton. Based on the available literature (see Table 1) and in cooperation with the German national coaches, the following hypotheses were formulated:

- (1) The rallies are shorter in the doubles than in the singles disciplines with the exception of the ladies' doubles.
- (2) The average resting times do not differ between the disciplines.
- (3) The ladies' doubles are characterized by the highest number of shots per rally.
- (4) The shots per second (frequency) reveal differences across all disciplines. The highest frequency is found in the men's doubles, followed by the mixed doubles, ladies' doubles, men's singles and, of lowest frequency, the ladies' singles.

Table 1. Overview of selected studies concerning the time structure in badminton (times in seconds [s])

Authors	Scoring system	Participants	Discipline	Rally time	Rest time	Shots.s ⁻¹
O'Donoghue, 1998	Traditional	European level	MS	6.7	11.3	0.95
			LS	5.7	9.5	0.82
			MD	5.3	10.9	1.25
			LD	5.7	9.5	1.15
			XD	4.8	9.8	1.20
Liddle & O'Donoghue, 1998	Traditional	European level	MS	9.15	13.84	-
			LS	6.73	11.03	
			MD	6.35	15.01	
			LD	7.61	11.72	
Cabello Manrique & González-Badillo, 2003	Traditional	International level	MS and LS	6.40	12.93	0.95
Cabello et al., 2004	Traditional	National junior-international level	MS and LS	8.1 (international level)	19.3 (international level)	-
Faude et al., 2007	Rally point	International level	Mean of MS and LS	5.5	11.4	0.92
Chen & Chen, 2008	Traditional and rally point (5x7)	National level	MS	7.9-8.2	-	0.95-1.02
Alcock & Cable, 2009	Traditional	National level	MS MD	5.0 5.0	8.1 9.7	0.9 1.2
Abián-Vicen et al., 2013	Rally point (3x21)	World Top Players	MS LS	9.05 7.95	24.65 17.90	1.09 0.92
Abian et al., 2014	Rally point (3x21)	World Top Players	MS	10.4	26.7	1.07

2. Methods

2.1. Sample

A total of 50 match videos of singles and doubles were selected according to the official world ranking of March 2011. The sample consisted of ten matches of the world's number 1-10 in each discipline ($n = 50$). All matches analysed took place between 2010 and 2012 in world-series tournaments. The matches were filmed and the video footage supplied by the German Badminton Federation or the Institute of Applied Sport Science in Germany (IAT).

2.2. Instruments

Analyses of the video files were undertaken using the software utilius®vs (version 4.5.2, ccc-software, Leipzig). For this software a complementary module for Badminton was developed by the IAT in collaboration with the German Badminton Federation, which was deployed for the systematic game analysis of badminton matches. Outside of this particular research the game analysis software was generally applied to record tactical parameters of world-class badminton matches in order to prepare the German national players for their opponents. In the current study the software was used to determine parameters of the time structure. The reliability of the system and method has been confirmed in an earlier study (Gawin, Beyer & Büsch, 2012).

2.3. Parameters

The recorded variables included the total match duration, the performance time as a percentage thereof, the rally time, the rest time, the number of shots per rally and the number of shots per second (shot.s^{-1}) as a specification of frequency.

2.4. Statistics

The differences between the disciplines were proved through one-way analysis of variance with the dependent factor discipline. The specific hypotheses (1), (3) and (4) made it possible to define planned contrasts. Therefore, in the analysis of the variance of these variables contrasts were defined in advance according to the hypotheses. As effect sizes either the η^2_{part} or r_{contrast} for the analysis of contrasts were determined. For the analysis of the parameter shots per rally, which is not interval-scaled data but represents frequency data, the Kruskal-Wallis test was executed and completed post hoc by the Mann-Whitney test.

3. Results

The average duration of a badminton match under the current scoring conditions in top level badminton ranges from 40-50 minutes (see Table 2). The longest average match duration was found in the men's singles while contrary to expectations the shortest match duration was found in the ladies' doubles. The differences between the disciplines in this parameter were not statistically significant ($F_{4,45} = 0.84$; $p = 0.51$). The highest value of the performance time of about 30% was observed in the women's disciplines. The performance time in the men's and mixed doubles was about ten percent lower. The average rally times ranged from 5.6-10.2 s in which the ladies' doubles consisted of the longest rallies and the mixed doubles the shortest (see Table 2).

Hypothesis (1) that the rallies will be shorter in the doubles than in the singles disciplines with the exception of the ladies' doubles was confirmed by the analysis of planned contrasts ($T[29.21] = 7.45$; $p < 0.001$; $r = 0.81$). The revealed differences were practically and statistically significant.

Table 2. Mean and standard deviation of the total match duration, value and range of the performance time and mean and standard deviation of the rally time

Discipline	Total match duration (h)	Performance time (%)	Rally time (s)
MS	0:49:54 $\pm$ 0:19:20	26.5 (21.4-31.7)	9.3 $\pm$ 1.5
MD	0:45:55 $\pm$ 0:16:27	20.4 (17.2-24.5)	6.7 $\pm$ 1.5
LS	0:47:28 $\pm$ 0:16:35	29.2 (24.1-32.9)	9.2 $\pm$ 1.4
LD	0:40:04 $\pm$ 0:10:40	30.1 (23.4-37.2)	10.1 $\pm$ 3.2
XD	0:40:33 $\pm$ 0:09:14	19.4 (17.1-21.5)	5.6 $\pm$ 0.5

Hypothesis (2) proposed that the average resting times would not differ between the disciplines. In fact, the rest times in the both men's disciplines (MS and MD) were slightly longer on average than in the women's disciplines and in the mixed doubles (see Table 3). The great variance in the ladies' doubles was conspicuous but the differences between the disciplines were not statistically significant ($F_{4.45} = 2.09$; $p = 0.10$). Therefore hypothesis (2) could also be confirmed.

In distinction to the approach followed by, for example, Abián-Vicen et al. (2013) and Abián et al. (2014), the next variables, namely the number of shots per rally and the shots per second, only refer to the shuttle contacts of one player or double pair and not to both sides. For the purpose of obtaining data of practical value for coaches, the research regarded the intensities of one side as sufficient and necessary. This has to be taken into consideration when comparing the results presented in this study with findings reported in similar publications. Specifically, the number of shots per second must be doubled to forge a useful comparison with other competition analyses.

Concerning the number of shots per rally, hypothesis (3) anticipates that the ladies' doubles manifests the highest number of shots in a rally. This hypothesis, however, could only be partially confirmed. Generally, most rallies in all five disciplines were finished after two shuttle contacts ($\text{modus} = 2$) of one player or doubles pair (see Table 3). The median was 4 shots per rally, except for the mixed doubles where the center of the distribution was located at 3 shots per rally. The proportion of rallies with a high number of shots appeared to be significantly higher in the ladies' doubles than in the men's doubles ($z = -3.0$; $p = 0.01$; $r_{DD-HD} = 0.08$), the ladies' singles ($z = -4.28$; $p < 0.01$; $r_{DD-DE} = 0.11$) and the mixed doubles ($z = -7.85$; $p < 0.01$; $r_{DD-MX} = 0.20$). But the effect size was small and the ladies' doubles did not significantly differ from the men's single in this parameter ($z = -1.55$; $p = 0.07$).

Table 3. Mean and standard deviation of the rest times, median, modus and range of the number of shots per rally and mean and standard deviation of the shots per second

Discipline	Rest time (s)	Shots per rally	Shots per second
MS	23.1 ± 3.9	4/2 (1-28)	0.56 ± 0.03
MD	23.3 ± 3.7	4/2 (1-34)	0.76 ± 0.03
LS	19.4 ± 3.4	4/2 (1-21)	0.49 ± 0.02
LD	20.0 ± 5.3	4/2 (1-41)	0.62 ± 0.05
XD	20.6 ± 3.2	3/2 (1-23)	0.72 ± 0.03

The lowest value of shots per second was observed in the ladies' singles, whereas the men's doubles followed by the mixed doubles turned out to be the 'fastest' disciplines (see Table 3). As stated in hypothesis (4) the differences between the disciplines were practically and statistically significant with a huge effect ($F_{4, 45} = 122.42$; $p < 0.001$; $\eta^2_{\text{part}} = 0.92$). The post hoc analysis revealed that only the men's doubles and mixed doubles showed comparable frequencies as expressed by the number of shots per second.

4. Discussion

Comparing the outcome of the present study to the research of Abián-Vicen et al. (2013) and Abián et al. (2014), a number of similarities stand out. In their studies, the authors listed average rally times from 9-10 s in the men's singles and from about 8 s in the ladies' singles. The rest times counted for 25-27 s in men's singles and 20 s in ladies' singles respectively. The matches of the male players were characterized by a frequency of 1.07-1.09 shots.s⁻¹, in the ladies' singles of 0.92 shots.s⁻¹. Reference values concerning these variables in the double disciplines under the current scoring conditions are not known.

The results of Abián and colleagues can be confirmed to a large extent by the present analysis given that only slight discrepancies emerge. For example, in this study the mean rally times in the ladies' singles lasted slightly longer at 9.2 s whilst the average resting times for the male players in the singles came up shorter at about 23 s. The frequency (shots.s⁻¹) measured here was higher in the men's singles (1.12 shots.s⁻¹) and in the ladies' singles (0.98 shots.s⁻¹ – to obtain these values it is necessary to double the frequency data of this study, as mentioned above) than in the studies of Abián and colleagues (2013; 2014). The divergences are likely due to the different framing conditions. Abián-Vicen et al. (2013) and Abián et al. (2014) analysed the finals games of the Olympic tournament 2012, while the present study is based on the matches of world-series tournaments. It should be noted that in the context of the Olympic Games it is probable that different constraints exist, one of which includes the players' awareness of the intense level of public interest.

As far as the doubles disciplines are concerned, to date no reference data are known regarding the analysed parameters. Therefore, the present study has focused primarily on the disciplines of ladies' doubles, men's doubles and mixed doubles. Especially in the ladies' doubles it was basically assumed that there exist other characteristics than in the men's or mixed doubles.

As expected, the performance times in the men's and mixed doubles showed a relatively low work to rest ratio of about 20 %. This value is higher in the singles disciplines with 26.5 % (MS) and 29.2 % (LS). It is related to the basic hypothesis that this ratio in the ladies' doubles with 30.1 % exceeds the values of the singles disciplines. A similar result could be found in the rally times. With average rally times of 10.1 s the longest rallies of all disciplines were measured in the ladies' doubles. In contrast to this outcome, the rest times in the ladies' doubles did not differ significantly from the other disciplines. This explains the high work to rest ratio in the ladies' doubles. The uniqueness of the ladies' doubles was also highlighted by the parameter shots per rally. The ladies' doubles consisted of significantly more rallies with a higher number of shuttle contacts than in the men's or the mixed doubles. The rally with the highest amount of shots per team (41) was measured in the ladies' double. On the strength of these results the basic assumption of this study that the discipline of elite level ladies' double differs from the characteristics of the two other double disciplines, men's doubles and mixed doubles, can be confirmed empirically.

5. Practical implications

Coaches of elite badminton, firstly, can make use of the obtained data by considering the *frequencies* (shots.s⁻¹) identified. If the badminton-specific endurance training is oriented to the performance of world-class badminton, then the on-court drills should be based on the discipline-specific frequencies. While executing "multi-shuttle" drills, common in top-level badminton, a frequency should be applied that in ladies' singles the exercising player has to hit a shuttle every 1.9-2.0 s. In the men's singles the corresponding frequency is 1.7-1.8 s, in ladies' doubles 1.5-1.6 s, and 1.2-1.4 s in the men's doubles and mixed doubles. It must be emphasized that these frequencies represent the average value for all types of shots. The frequency has to be set higher if practicing fast shots in the fore- or mid-court and it has to be adjusted downwards when performing drills with shots from the rear-court that require the player to cover a longer distance.

Secondly, the training *volumes* should correspond to discipline-specific requirements. Besides the most common work intervals of 6-10 s the highest possible values should also be applied in training drills. For example in the ladies' doubles this means that maximal rally durations of up to 80 s are possible. This, too, should be covered in the badminton-specific endurance training on court. Finally, it should be noted that these recommendations were discussed with the German national coaches and have subsequently been incorporated into the training regimes of the national top players.

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